


Plastic Degradation Hot Paper

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Tandem Catalysis for Plastic Depolymerization: *In Situ* Hydrogen Generation via Methanol Aqueous Phase Reforming for Sustainable Polyethylene Hydrogenolysis

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Abstract: Depolymerizing plastic waste through hydrogen-based processes, such as hydrogenolysis and hydrocracking, presents a promising solution for converting plastics into liquid fuels. However, conventional hydrogen production methods rely heavily on fossil fuels, exacerbating global warming. This study introduces a novel approach to plastic waste hydrogenolysis that utilizes *in situ* hydrogen generated via the aqueous phase reforming (APR) of methanol, a biomass-derived chemical offering a more sustainable alternative. Our results show that a bimetallic Ru–Pt/TiO₂ catalyst achieved high conversion (85.1 %) and selectivity (81.0 %) towards liquid fuels and lubricant oils in a tandem process combining polyethylene (PE) hydrogenolysis and methanol APR. By tuning the metal loading, we identified that Pt enhances hydrogen production through methanol APR, while Ru drives C–C bond cleavage, which is crucial for PE hydrogenolysis. Isotope labeling analysis confirmed that hydrogen generated from methanol APR is effectively utilized in the PE hydrogenolysis reaction. This method was also successfully applied to post-consumer polyolefin waste, with selectivity toward valuable products ranging from 75.0 % to 88.9 %. This study highlights an innovative strategy to reduce reliance on fossil-fuel-derived hydrogen in plastic waste depolymerization, promoting both sustainability and environmental protection.

Introduction

Plastics are indispensable in modern life due to their durability, convenience, and large-scale manufacturability.^[1] Global plastic production reached over 413.8 million tonnes in 2023, driven by growing consumer demand across various sectors, such as food packaging automotive, construction, and industrial components.^[2] However, the majority of plastic waste is either incinerated (19 %) or sent to landfills (49 %),^[3] leading to serious environmental concerns. Current recycling efforts are dominated by mechanical recycling, which is cost-effective but results in a loss of plastic quality with each cycle, limiting its reuse potential.^[4] Chemical recycling, on the other hand, offers the ability to convert plastic waste into valuable chemicals or monomers, enabling more sustainable reuse.^[5] Among chemical recycling methods, pyrolysis is widely studied, but its high reaction temperature requirements (400–900 °C) and broad product distribution complicate downstream separation, reducing overall efficiency.^[6]

Hydrogenolysis has emerged as a promising alternative for plastic recycling, operating at lower temperatures (200–350 °C) and offering high selectivity toward valuable liquid fuels and waxes.^[7] This process is particularly effective for polyethylene (PE) and polypropylene (PP), which account for more than half of global plastic production due to their short lifespan and extensive use.^[6c,8] Despite its advantages, hydrogenolysis generally requires high-pressure hydrogen to cleave C–C bonds in polyolefins, a process that relies heavily on hydrogen produced from fossil fuels.^[7b,9] The carbon-intensive nature of fossil-fuel-derived hydrogen significantly detracts from the environmental benefits of plastic depolymerization, underscoring the need for greener hydrogen sources.^[9e,10]

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Several alternative approaches have been explored to reduce reliance on external hydrogen gas in plastics depolymerization.^[9b,11] Promising results have been obtained using catalysts such as Pt/ γ -Al₂O₃,^[11a] Pt-Fe/Al₂O₃,^[11b] and Ru/HZSM-5,^[9b] which enable PE conversion without an external hydrogen supply. However, these processes require a large quantity of catalyst relative to the amount of plastic converted^[11] and typically generate complex product mixtures, including alkanes, alkenes, aromatics, and cycloalkanes, which complicate separation and reduce overall efficiency.^[9b,11–12] In particular, separating alkanes from aromatics poses significant challenges due to their similar boiling points and azeotrope formation, which limits the practicality of liquid fuels derived from these processes.^[13] Liquid organic hydrogen carriers (LOHCs) have also been investigated, but issues like catalyst poisoning by dehydrogenation products (e.g., naphthalene) hinder the efficiency of these systems.^[14]

In response to the growing demand for greener environmentally friendly hydrogen sources, green hydrogen – produced via electrolysis powered by renewable energy – has gained significant attention. Despite its potential to drastically reduce carbon emissions, less than 5 % of global hydrogen is currently produced through sustainable electrolysis methods.^[15] Meanwhile, biomass has emerged as a versatile renewable resource with significant potential to contribute to decarbonization efforts.^[16] For instance, biofuels derived from biomass are projected to replace up to 27 % of global transportation fuel by 2050, highlighting the importance of biomass in sustainable energy systems.^[15a,17]

Beyond biofuels, biomass can also serve as a feedstock for hydrogen production, providing an alternative pathway to complement electrolysis, particularly in regions or applications where renewable energy access is constrained.^[18] Aqueous phase reforming (APR) of biomass-derived feedstocks – such as methanol, ethanol, and ethylene glycol – represents a promising alternative for green hydrogen generation.^[15a,19] Methanol, typically produced from biomass via gasification, faces high energy demands due to the elevated reaction temperatures required.^[20] However, advancements like chemical looping can reduce these demands while inherently lowering carbon dioxide emissions.^[21] Due to its mild operation temperatures (200–350 °C), APR can be coupled with PE hydrogenolysis, enabling in situ carbon-neutral hydrogen production.^[15a] This combination offers a promising pathway for improving both the environmental and economic aspects of plastic recycling.

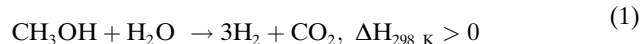
In this study, we developed a tandem process that integrates PE hydrogenolysis with methanol APR. Among various catalysts, the bimetallic Ru–Pt/TiO₂ catalyst exhibited the highest activity in the tandem PE hydrogenolysis/methanol APR process. To elucidate the relationship between catalyst properties and reactivity, we employed a range of analytical techniques, including CO chemisorption, Scanning Transmission Electron Microscopy - Energy Dispersive X-ray Spectroscopy (STEM-EDS) mapping, X-ray Photoelectron Spectroscopy (XPS), Hydrogen Temperature Programmed Reduction (H₂-TPR), Mass Spectrometry

(MS) and Attenuated Total Reflection - Fourier Transform Infrared Spectroscopy (ATR-FTIR). Our findings revealed that Ru plays a crucial role in enhancing C–C bond cleavage, while Pt effectively produces hydrogen from the methanol APR process. Mass spectrometry and ATR-FTIR analyses confirmed the effective utilization of in situ generated hydrogen from methanol APR in the PE hydrogenolysis reaction. Furthermore, the Ru–Pt/TiO₂ catalyst demonstrated excellent reusability and high conversion (71.6–85.3 %) across various post-consumer plastics, including LDPE bags, HDPE bottles, and commercial LDPE. This work provides valuable insights into the behavior of Ru–Pt catalysts in tandem reactions and offers a novel, more sustainable route for plastic depolymerization, paving the way for future advancements in green recycling technologies.

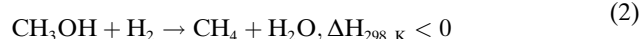
Results and Discussion

PE hydrogenolysis was conducted using a methanol-water mixture without the addition of external hydrogen gas. Previous studies have demonstrated that Pt catalysts are effective in both APR^[22] and PE hydrogenolysis.^[11a,23] To explore this further, Pt catalysts supported on TiO₂, SiO₂, and γ -Al₂O₃ were synthesized and evaluated for their reactivity in PE hydrogenolysis using the methanol-water mixture. In this process, methanol and water react to produce H₂ and CO₂ in a stoichiometric ratio of 3:1 (Eq.1). Subsequently, methanol can react with H₂ to form CH₄ and H₂O (Eq.2).^[24] Additionally, CO₂ can react with H₂ either through methanation to produce CH₄ and H₂O (Eq.3) or via the reverse water-gas shift reaction to yield CO and H₂O (Eq.4), with both reactions being reversible.^[24] Since GC analysis confirmed the absence of CO (Figures S1–S4), only three reactions (Eq.1–3) were considered.

Methanol APR :



Methanol reduction :



CO₂methanation :



Reverse water gas shift :



While the Pt catalysts facilitated these reactions, only a limited amount of the generated hydrogen was used for PE hydrogenolysis, resulting in a low PE conversion of 6.6 % to 10.5 %, regardless of the support material (Figure 1a). This low PE conversion likely stems from insufficient hydrogen involvement in cleaving the C–C bonds in PE.^[25] To address this, Ru catalysts – known for their superior activity in C–C bond cleavage – were tested.^[11b,7b] The Ru catalysts exhibited slightly higher PE conversion, ranging from 16.6 % to 29.7 %, primarily due to enhanced hydrogen consumption.

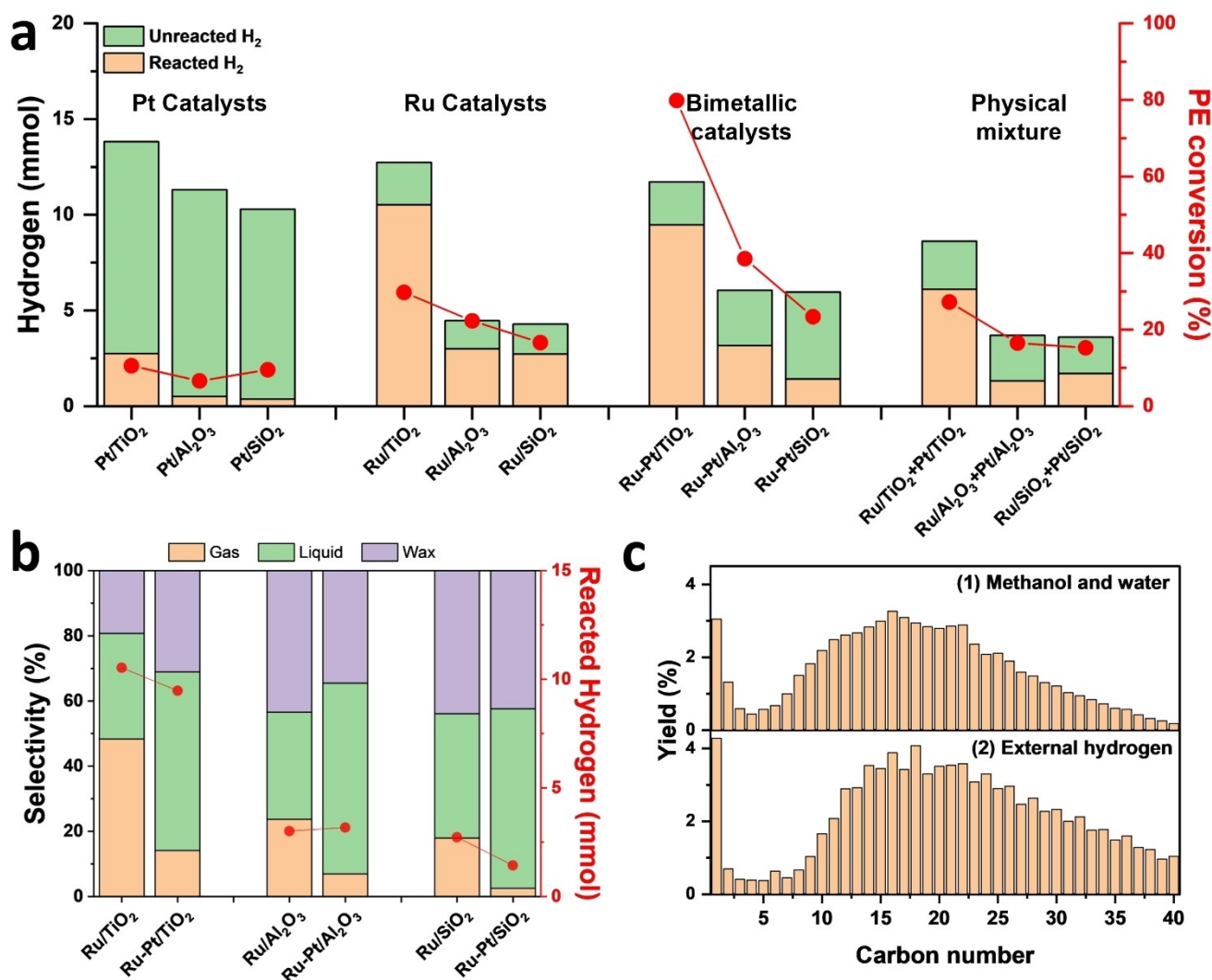


Figure 1. PE hydrogenolysis reaction with in situ generated hydrogen via methanol APR using different supported metal catalysts. a. Reacted and unreacted hydrogen, and PE conversion with different supported metal catalysts. b. Product selectivity and the amount of hydrogen reacted with PE on Ru and Ru–Pt catalysts. c. (1) Product distribution of the tandem PE hydrogenolysis/methanol APR reaction using Ru–Pt/TiO₂. Reaction conditions: PE (M_n : ~1,700, M_w : ~4,000) 1 g, methanol 0.2 g, water 1.8 g, catalyst 0.1 g, 325 °C, 500 rpm, 24 h, 1 MPa Ar. Detailed results are shown in Table S1. (2) Product distribution of PE hydrogenolysis using Ru/SiO₂ catalyst, known to be active for the reaction. Reaction conditions: PE (M_n : ~1,700, M_w : ~4,000) 1.7 g, catalyst 0.05 g, 250 °C, 500 rpm, 3 h, 3 MPa H₂. “Reacted hydrogen” refers to the hydrogen consumed during the PE hydrogenolysis reaction, while “unreacted hydrogen” denotes the hydrogen remaining in the reactor after the reaction. The detailed calculation methods for both are provided in the Supporting Information.

However, increased methane production from methanol side reactions reduced the overall PE conversion, following the trend: Ru/TiO₂ > Ru/ γ -Al₂O₃ > Ru/SiO₂ (Table S1). This suggests that hydrogen was diverted to side reactions, such as methanol reduction^[26] or CO₂ methanation,^[27] limiting its availability for PE hydrogenolysis. Although Ru/TiO₂ exhibited higher conversion (29.7 %) than other Ru-supported catalysts, substantial methane production indicated that hydrogen was primarily used for terminal C–C bond cleavage, resulting in high methane yields and solid residue (Table S1). Therefore, it is crucial to minimize hydrogen consumption in side reactions and direct the hydrogen toward internal C–C bond cleavage for improved PE hydrogenolysis performance.

To address these limitations, Ru–Pt bimetallic catalysts, known to promote PE hydrogenolysis^[1b] and APR,^[28] were employed. These bimetallic catalysts significantly outperformed the monometallic Pt and Ru catalysts, achieving PE conversions of 23.4 % to 79.9 %. The Ru–Pt bimetallic catalysts demonstrated more effective hydrogen utilization in PE hydrogenolysis than Pt catalysts and required less hydrogen than Ru catalysts, despite the higher conversion. Unlike the tandem PE hydrogenolysis/methanol APR process, where the Ru–Pt/TiO₂ catalyst is most effective without an external hydrogen supply, the monometallic Ru catalyst is the most active under conventional hydrogenolysis conditions with external hydrogen, consistent with previous literature^[9a,29] (Table S2). A comparison of selectivity be-

tween gas, liquid, and wax products (Figure 1b) showed that Ru catalysts favored gas production, including methane, while Ru–Pt catalysts exhibited higher selectivity for liquid products.

These findings align with previous reports, suggesting that the Ru–Pt bimetallic catalyst suppresses terminal C–C bond cleavage while promoting internal C–C bond cleavage, likely due to its preference for internal C–H bond activation by forming C–M (M = metal) bonds, thereby cleaving the corresponding C–C bond.^[1b] Among the Ru–Pt bimetallic catalysts, Ru–Pt/TiO₂ exhibited the highest conversion (79.9 %) and selectivity for liquid fuels such as gasoline, diesel, and lubricant oil (82.8 %), underscoring the effectiveness of this tandem catalytic process. At similar conversion levels (Table S1), the carbon distribution analysis (Figure 1c) for the Ru–Pt/TiO₂ catalyst using in situ generated hydrogen was similar to that obtained under conventional hydrogenolysis conditions with external hydrogen. GC-FID, GC/TOF-MS, and ¹³C NMR results (Figure S5–S7) confirmed that the products predominantly consisted of *n*-alkanes, simplifying the downstream separation process.^[30]

A physical mixture of Ru and Pt catalysts in a 1:1 ratio was tested to examine the importance of proximity between Ru and Pt species. This configuration resulted in PE conversions comparable to or lower than Ru monometallic catalysts (15.2 % - 27.2 %). While the amount of hydrogen participating in the PE reaction was higher than with the Pt catalyst alone, it was still lower than with the Ru catalyst alone (Table S1). This reduced hydrogen reactivity is likely due to the lower proportion of Ru, which limited hydrogen consumption for C–C bond cleavage in PE. Consequently, the physical mixture exhibited substantially lower PE conversion than the bimetallic catalyst, underscoring the critical role of intimate Ru–Pt interactions for efficient tandem PE hydrogenolysis/methanol APR. Elemental mapping confirmed a spatial overlap of Ru and Pt in the Ru–Pt/TiO₂ catalyst, suggesting that the proximity of the two metals contributes to the enhanced performance of the bimetallic system (Figure 2 and Figure S8). The superior performance of the Ru–Pt bimetallic catalysts, particularly on TiO₂, warrants further investigation into the specific factors responsible for their improved activity. CO-chemisorption analysis revealed similar particle sizes across all bimetallic catalysts (Table S3), indicating that factors other than particle size contribute to enhanced reactivity.

To explore the electronic structure of Ru and Pt in the bimetallic catalysts, X-ray photoelectron spectroscopy (XPS) analysis was conducted. The XPS spectra for Pt 4f and Ru 3d were deconvoluted into two pairs of doublets (Figure 3a, b). Due to the overlap between Ti 2p and Ru 3p, the Ru 3d spectrum was deconvoluted from the C 1s peak.^[31] Distinct shifts in the binding energies were observed between the monometallic and bimetallic catalysts: a 0.1–0.5 eV increase for Pt and a 0.2 eV decrease for Ru in the bimetallic catalyst (Table S4). These shifts are consistent with prior theoretical predictions of a 0.35 eV shift in the Pt 4f_{7/2} core level for a Pt–Ru alloy (60:40 molar ratio),^[32] and an experimental shift of 0.39 eV was also reported in the same study.^[32] This alignment suggests electron transfer

between Pt and Ru in the bimetallic system, accompanied by rehybridization of both the *d*-band and *sp*-band, as previously noted.^[32]

Given that active metal sites often exhibit different behaviors in their metallic versus oxidized states, we calculated the ratio of various Pt oxidation states for both the monometallic and bimetallic catalysts. It is well established that metallic Pt (Pt⁰) is crucial for methanol APR.^[33] In this study, the bimetallic catalyst exhibited a substantially higher Pt⁰ content (74.0 %) than the monometallic catalyst (57.2 %). This higher metallic Pt content was reflected in the hydrogen production, where the bimetallic catalyst produced 4.7 mmol_{H₂}/mg_{Pt}, surpassing the 2.8 mmol_{H₂}/mg_{Pt} generated by the Pt monometallic catalyst. On the other hand, Ruⁿ⁺ species are known to promote C–C bond dissociation, while Ru⁰ facilitates C–H activation during PE hydrogenolysis.^[34] The proportion of Ru⁴⁺ species was higher in Ru–Pt/TiO₂, reaching 40.6 %, compared to 26.2 % in Ru/TiO₂. Conversely, the content of metallic Ru species was more prevalent in Ru/TiO₂ (73.8 %) than in Ru–Pt/TiO₂ (59.4 %). Notably, the PE conversion was noticeably different – 29.7 % for the Ru monometallic catalyst and 79.9 % for the bimetallic catalyst.

As the oxidation states of metal species play a crucial role in catalytic activity, H₂ temperature-programmed reduction (H₂-TPR) was conducted to investigate the differences between monometallic and bimetallic catalysts on TiO₂ (Figure 3c). For Ru/TiO₂, two reduction peaks were observed between 100 °C and 200 °C, corresponding to the reduction of Ru oxides to Ru⁰,^[35] likely due to variations in Ru particle size.^[35] Pt reduction occurred at 102.2 °C, attributed to the reduction of Pt oxide.^[36] Due to the different reduction degrees of Ru (Ru⁴⁺ to Ru⁰) and Pt (Pt²⁺ to Pt⁰), hydrogen consumption per equivalent of metal reduced was calculated for each metal. The hydrogen consumption for metal reduction followed the trend: Ru/TiO₂ (0.523 mmol-H₂/equivalent of metal reduced) > Ru–Pt/TiO₂ (0.276 mmol-H₂/equivalent of metal reduced) > Pt/TiO₂ (0.014 mmol-H₂/equivalent of metal reduced) (Table S5). The higher hydrogen consumption for Ru/TiO₂ compared to Pt/TiO₂ can be attributed to the higher reducibility of Ru species relative to Pt species.^[37] To further evaluate the reducibility of the catalysts based on their support materials, additional H₂-TPR analysis was performed (Figure 3d). All bimetallic catalysts exhibited a reduction peak between 100–200 °C, indicating the reduction of both Ru oxides and Pt oxides.^[35–36,38] The single peak in the TPR spectra for the Ru–Pt bimetallic catalysts suggests the formation of Ru–Pt bimetallic interactions due to the proximity of Pt and Ru species.^[39] The hydrogen consumption was measured as 0.545 mmol-H₂/equivalent of metal reduced for Ru–Pt/SiO₂, 0.503 mmol-H₂/equivalent of metal reduced for Ru–Pt/γ-Al₂O₃, and 0.276 mmol-H₂/equivalent of metal reduced for Ru–Pt/TiO₂ (Table S5). The increasing PE conversion with decreasing hydrogen consumption may be associated with the highest proportion of Ruⁿ⁺ species in Ru–Pt/TiO₂ among the bimetallic catalysts, consistent with the XPS analysis results.

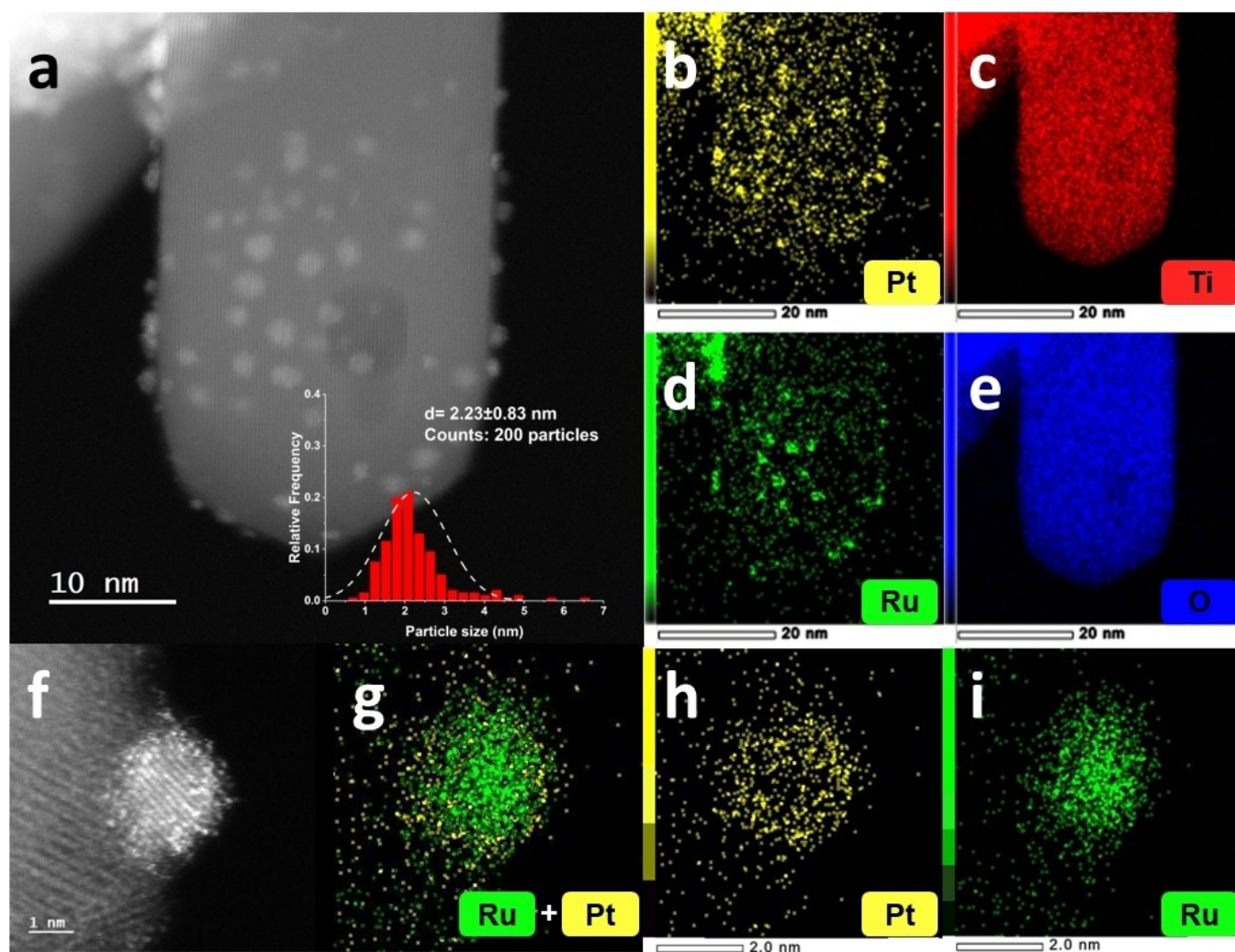


Figure 2. a. HAADF-STEM image and metal particle size distribution measured by STEM images. b-e. Corresponding EDS mappings of Pt, Ti, Ru, and O elements in Ru–Pt/TiO₂ catalysts, respectively. f. HAADF-STEM image of a metal particle at a resolution of 1 nm. g. Corresponding EDS mapping showing the overlap of Ru and Pt. h-i. Individual EDS mappings for Pt and Ru, respectively.

To deepen our understanding of the tandem PE hydrogenolysis/methanol APR, a series of experiments were conducted using Ru–Pt/TiO₂ catalysts, focusing on the effects of reaction temperature and time (Table S6 and S7). PE conversion increased from 44.0 % to 84.4 % as the reaction time extended from 6 to 36 hours (Table S6). Notably, beyond 12 hours, the conversion plateaued, showing minimal further increase despite the prolonged reaction time. Reaction temperature had a more pronounced impact, with PE conversion improving from 16.7 % at 275 °C to 97.8 % at 350 °C, accompanied by a corresponding rise in hydrogen consumption (Table S7). Additionally, methane production increased proportionally with PE conversion (Table S7). The amount of methane derived from PE can be calculated using measurable variables, as detailed in the Supporting Information. Subtracting this value from the total measured methane provides the amount generated from methanol. A closer analysis revealed that while methane produced from PE increased with temperature,

methane generated from methanol decreased (Table S7). This behavior can be attributed to the thermodynamics of the involved reactions. Methanol APR, an endothermic process (Eq. 1), becomes more favorable at higher temperatures, suppressing methane formation from methanol. In contrast, methanol reduction (Eq. 2) and CO₂ methanation (Eq. 3) are exothermic, leading to decreased methanol-derived methane at elevated temperatures (Table S7).^[40]

Furthermore, experiments with varying water/methanol ratios were conducted to explore the individual roles of methanol and water in the tandem reaction. Under supercritical water conditions (temperature > 374 °C, pressure > 22 MPa), water exhibited a dielectric constant comparable to that of PE, promoting PE solubility and reactivity.^[41] In contrast, under subcritical conditions, water becomes denser and more polar, which inhibits PE hydrogenolysis reactivity.^[42] Consequently, in the presence of water alone at 325 °C (subcritical conditions), the conversion was low, at 7.4 % (Figure 4a). Methanol, with its lower dielectric con-

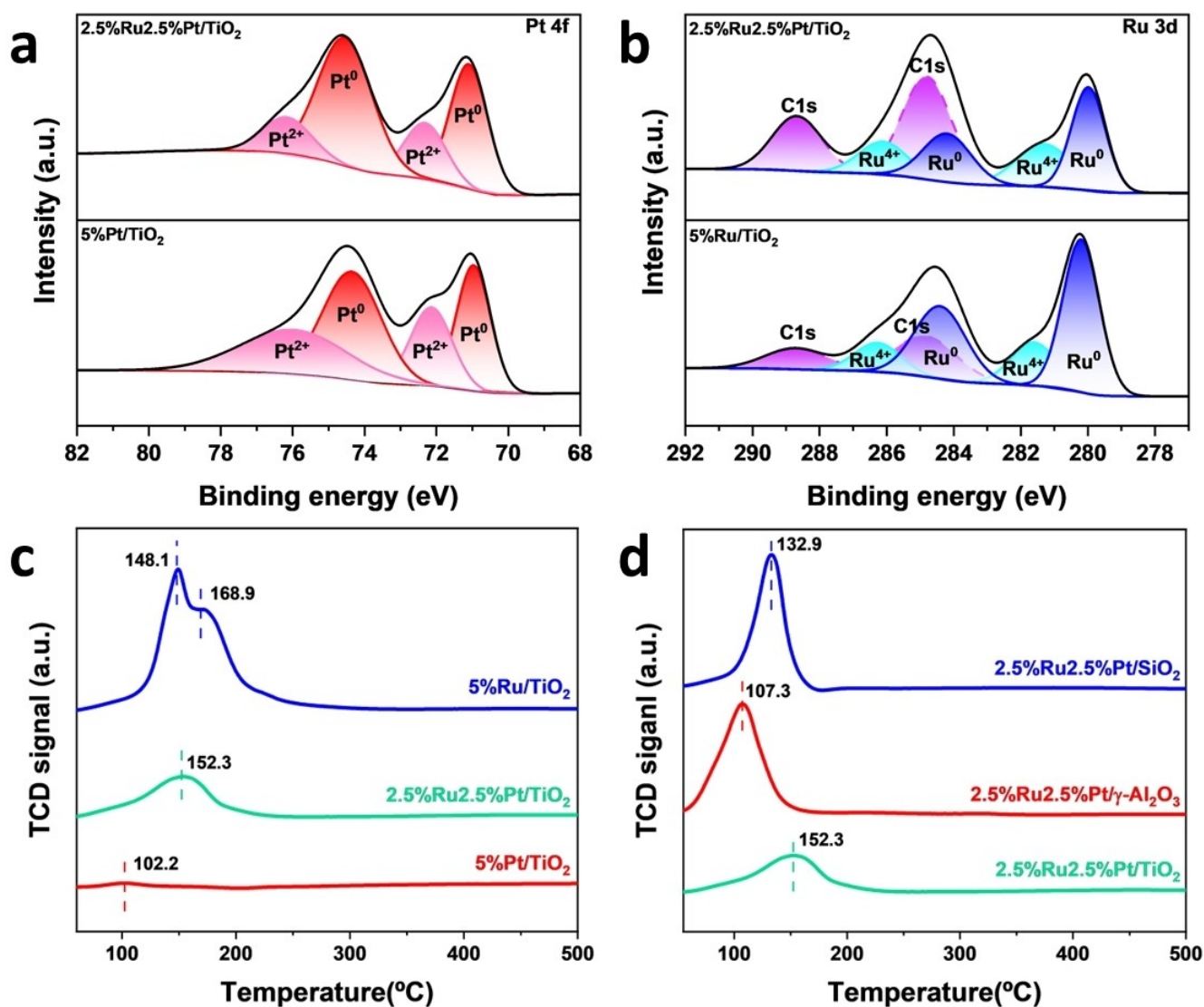


Figure 3. a. Pt 4f XPS spectra of Pt/TiO₂ and Ru–Pt/TiO₂ catalysts. b. Ru 3d XPS spectra of Ru/TiO₂ and Ru–Pt/TiO₂ catalysts. c. H₂-TPR with catalysts supported on TiO₂. d. H₂-TPR with Ru–Pt catalysts supported on various metal oxides.

stant compared to water, has been reported to improve reactivity in PE hydrothermal and solvothermal liquefaction.^[43] Using methanol alone increased PE conversion to 56.8 %, significantly higher than with water alone. Conversion also varied with the methanol-to-water ratio, as the methanol APR reaction requires a 1:1 molar ratio for optimal hydrogen production (Eq.1). When the methanol fraction increased to 25 %, the highest conversion of 85.1 % was achieved, highlighting the effectiveness of the tandem PE hydrogenolysis/methanol APR process (Figure 4a and Table S8). In order to confirm that mass transport limitation under the optimized conditions is negligible, we measured the activity of the 2.5 %Ru-2.5 %Pt/TiO₂ catalyst by varying stirrer speeds.^[44] We found that at 300 rpm, a slightly lower conversion of 74.7 % was observed (Table S9), but conversion consistently ranged from 84.0 % to 85.1 % at 400 to 600 rpm (Table S9). These results confirm that 500 rpm

ensures optimal mixing of the two distinct phases, water/methanol and reactants.

To elucidate the specific roles of Ru and Pt in bimetallic catalysts, a series of TiO₂-supported bimetallic catalysts with a total loading of 5 % was synthesized, varying the proportions of Ru and Pt. The monometallic Pt/TiO₂ catalyst showed a low conversion of 10.8 % with limited hydrogen consumption (0.6 mmol) for PE hydrogenolysis (Figure 4b). However, increasing the Ru loading to 1 % and 2.5 % further enhanced hydrogen consumption (7.2 and 12.4 mmol, respectively), resulting in higher PE conversions of 71.7 % and 85.1 %, respectively. When the Ru loading was increased to 4 %, the conversion decreased to 77.2 %, likely due to a reduced amount of hydrogen available for PE hydrogenolysis (Figure 4b). This trend of reduced hydrogen availability with increasing Ru content (Table S10) suggests that the lower Pt content limits the hydrogen production from methanol APR.^[29c,45] Furthermore, excess Ru promotes

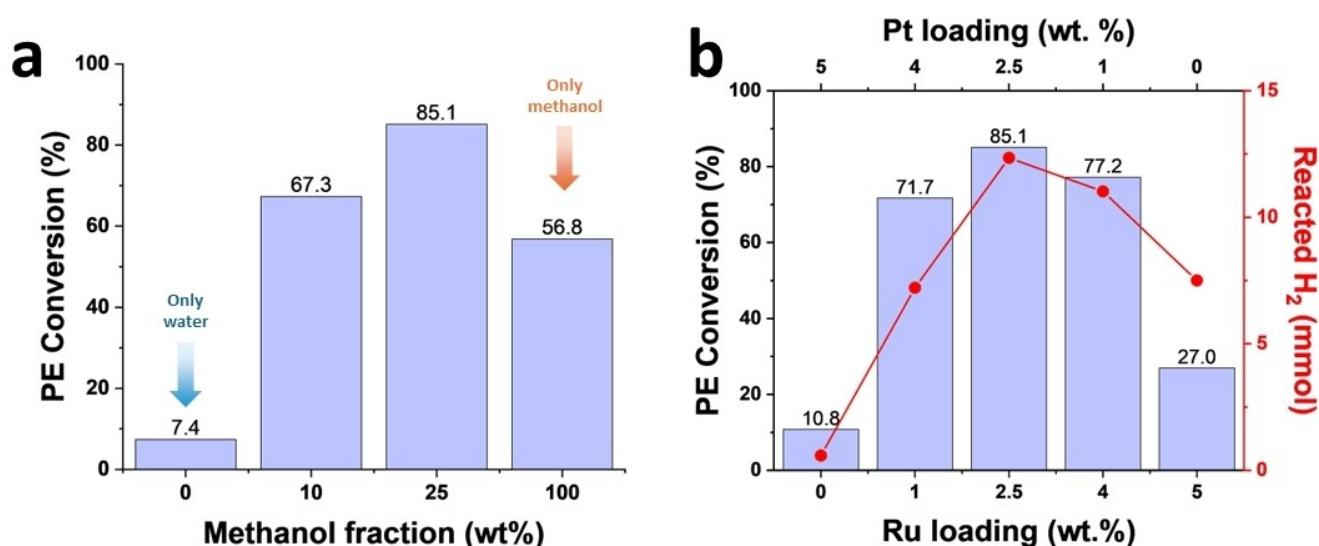


Figure 4. PE conversion and produced methane as a function of reaction temperature or methanol fraction. a. PE conversion as a function of methanol fraction. Reaction conditions: PE (M_n : ~1,700, M_w : ~4,000) 1 g, 2.5%Ru-2.5%Pt/TiO₂ catalyst 0.1 g, 325 °C, 500 rpm, 12 h, 1 MPa Ar. Detailed results are shown in Table S8. b. PE conversion and reacted hydrogen as a function of metal loading of Ru–Pt/TiO₂ catalysts. Reaction conditions: PE (M_n : ~1,700, M_w : ~4,000) 1 g, catalyst 0.1 g, 325 °C, 500 rpm, 12 h, 1 MPa Ar, and methanol 0.5 g and water 1.5 g. Detailed results are shown in the Table S10.

hydrogen consumption through methanation,^[46] further reducing the hydrogen available for PE hydrogenolysis. Interestingly, the monometallic Ru/TiO₂ catalyst consumed a similar amount of hydrogen as 1%Ru-4%Pt/TiO₂ but achieved a much lower PE conversion of only 27.0%, compared to 71.7% for 1%Ru-4%Pt/TiO₂ (Figure 4b). Specifically, methane production from PE was significantly higher for Ru/TiO₂ at 6.50 mmol, compared to 3.52 mmol for 1%Ru-4%Pt/TiO₂ (Table S10), suggesting that Ru/TiO₂ primarily utilized the hydrogen from methanol APR for terminal C–C bond cleavage. This led to the formation of more solid residues and methane, as previously discussed in Figure 1a. In contrast, the optimal incorporation of Pt appears to shift the cleavage preference toward internal C–C bonds,^[1b] enhancing overall PE conversion. Therefore, achieving an optimal balance between Ru and Pt is crucial to maximizing hydrogen efficiency and ensuring the effective progression of both methanol APR and PE hydrogenolysis.

ATR-FTIR analysis was conducted using deuterated substrates and n-dodecane to confirm that the hydrogen used in PE hydrogenolysis originated from methanol APR. Four experimental conditions were evaluated: 1) H₂O/CH₃OH, 2) D₂O/CH₃OH, 3) H₂O/CD₃OD, and 4) D₂O/CD₃OD. Methanol was used as the limiting reactant in all experiments, and GC analysis confirmed that methanol was fully consumed in each case (Figure S9). The liquid products from all four reactions exhibited peaks at 2920 cm⁻¹ and 2855 cm⁻¹, corresponding to CH₃ and CH₂ stretching, respectively (Figure 5a).^[47] In reactions involving D₂O or CD₃OD, a peak at 2146 cm⁻¹, corresponding to aliphatic C–D stretching, was observed (Figure 5a, b), while no such peak was detected in reactions with H₂O and CH₃OH. The C–D stretching peak indicates that deuterium was incorpo-

rated into the products when CD₃OD or D₂O was used. To quantify the extent of deuterium incorporation, the relative areas of the C–D and C–H peaks were integrated. The ratios of C–D peak area to C–H peak area for the H₂O/CD₃OD, D₂O/CH₃OH, and D₂O/CD₃OD systems were 0.03, 0.08, and 0.14, respectively. Correspondingly, the ratios of deuterium incorporation in these reactions were 1:2.8:4, respectively. The theoretical ratios of available deuterium in the H₂O/CD₃OD, D₂O/CH₃OH, and D₂O/CD₃OD systems were calculated to be 1:2.7:3.7, closely aligning with the experimental results. These findings indicate that hydrogen produced from methanol APR was used in the formation of liquid products.

To understand the mechanism of tandem PE hydrogenolysis/methanol APR in gas product formation, experiments were conducted using deuterium-labeled methanol and water. The gas products were analyzed by mass spectrometry, as shown in Figure 5c, d. Four experimental conditions, identical to those used in ATR-FTIR analysis, were evaluated. Methanol APR was initially performed without PE (Figure 5c). The distribution of hydrogen species produced during the reaction was investigated, with $m/z=2$ corresponding to H₂, $m/z=3$ to HD, and $m/z=4$ to D₂ (Figure 5c). In the H₂O/CH₃OH system, where H is the only hydrogen source present, the dominant peak at $m/z=2$ indicated the production of H₂. Likewise, in the D₂O/CD₃OD system, where D is the only hydrogen source present, the dominant peak at $m/z=4$ indicated the formation of D₂. However, in the H₂O/CD₃OD and D₂O/CH₃OH cases, peaks at $m/z=2$, 3, and 4 were observed due to the presence of both H and D in the hydrogen source. The $m/z=3$ peak for HD suggests that deuterium isotope exchange occurs between water and methanol during the

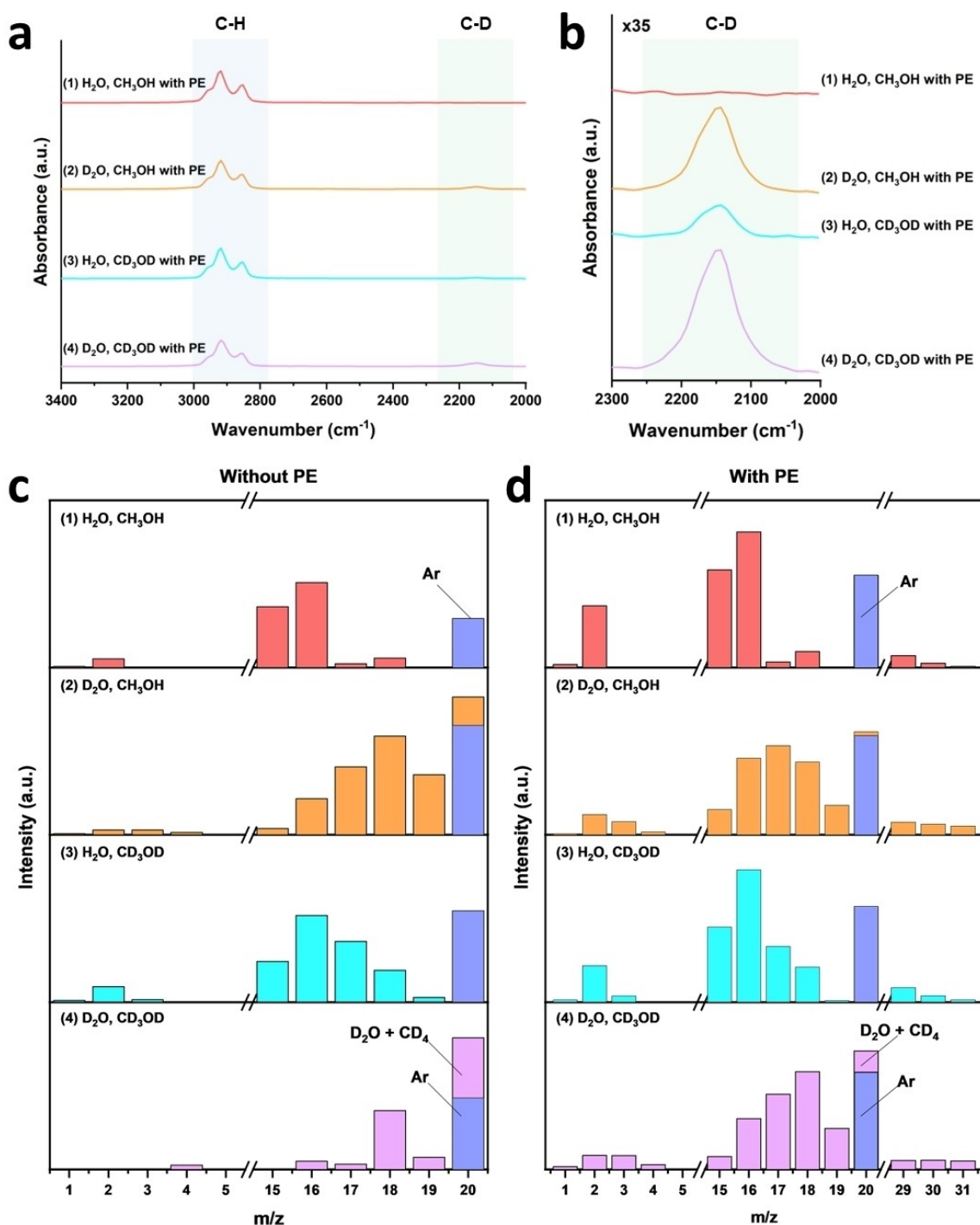


Figure 5. a. ATR-FTIR spectra of liquid products after reaction with (1) n-dodecane and H₂O and CH₃OH (2) n-dodecane and D₂O and CH₃OH (3) n-dodecane and H₂O and CD₃OD (4) n-dodecane and D₂O and CD₃OD. b. C–D vibration range scaled up 35 times. Reaction conditions: n-dodecane 1 g, Ru–Pt/TiO₂ catalyst 0.1 g, 325 °C, 500 rpm, 12 h, 1 MPa Ar, and CH₃OH or CD₃OD 0.1 g, and H₂O or D₂O 0.3 g. c. mass spectrometry (m/z = 1–20) with methanol reforming: (1) using H₂O and CH₃OH (2) using D₂O and CH₃OH (3) using H₂O and CD₃OD (4) using D₂O and CD₃OD. d. mass spectrometry (m/z = 1–20 and 29–31) with tandem PE hydrogenolysis/methanol APR (1) using H₂O and CH₃OH (2) using D₂O and CH₃OH (3) using H₂O and CD₃OD (4) using D₂O and CD₃OD. Reaction conditions: PE (*M*_n: ~1,700, *M*_w: ~4,000) 1 g, Ru–Pt/TiO₂ catalyst 0.1 g, 325 °C, 500 rpm, 12 h, 1 MPa Ar, and CH₃OH or CD₃OD 0.5 g, and H₂O or D₂O 1.5 g.

reaction, resulting in the formation of CH₃OD or CD₃OH and the production of HD.^[48] To confirm whether this hydrogen was utilized in PE hydrogenolysis, methanol APR experiments were conducted in the presence of PE (Fig-

ure 5d). In both cases, the intensity of the m/z = 2 peak increased compared to the methanol APR without PE. Notably, when D₂O/CD₃OD were used, only the peak at m/z = 4 was observed without PE; however, with the

presence of PE, peaks at $m/z=2$, 3, and 4 were detected, confirming the production of H_2 , HD, and D_2 . This suggests that H_2 was generated during the dehydrogenation process involved in PE hydrogenolysis.^[1b,23c,49]

To confirm that the hydrogen produced in the methanol APR was indeed utilized for PE hydrogenolysis, we focused on analyzing peaks in the methane range ($m/z=15-20$) and those in the ethane range ($m/z=29-31$). Under the reaction conditions, methane can be produced either from methanol (Eq. 2, 3) or PE hydrogenolysis. Therefore, the presence of deuterated methane complicates the determination of its origin. Consequently, we focused on the peaks in the ethane range to determine whether the hydrogen produced during APR was used in the PE hydrogenolysis reaction. The $m/z=29$ peak primarily represents fragment ions of C_2H_6 , deuterated C_2H_6 , C_3H_8 , deuterated C_3H_8 and CH_3OH/CH_3OD (Table S12–14), while $m/z=30$ corresponds to fragment ions of C_2H_6 , deuterated C_2H_6 , deuterated C_3H_8 , CD_3OH/CD_3OD (Table S12–14). The $m/z=31$ peak is associated with deuterated C_2H_6 , deuterated C_3H_8 , and CH_3OH (Table S12–14). In comparing the reactions using H_2O/CH_3OH and D_2O/CH_3OH (Figure 5d), the D_2O/CH_3OH reaction showed an increase in peak intensity at $m/z=30$ compared to the H_2O/CH_3OH reaction. Given that D_2O/CH_3OH primarily produces CH_3OD rather than CD_3OH ,^[48] this increase in $m/z=30$ intensity can be attributed to the formation of deuterated hydrocarbons (C_2H_6 and C_3H_8). Meanwhile, the $m/z=31$ peak showed higher intensity in the reactions with D_2O/CD_3OD compared to H_2O/CH_3OH , suggesting increased formation of deuterated C_2H_6 and deuterated C_3H_8 . These findings confirm that the hydrogen generated during methanol APR was used in PE hydrogenolysis.

Reusability experiments were conducted to evaluate the recyclability of the Ru–Pt/TiO₂ catalyst in the tandem PE hydrogenolysis/methanol APR reaction. The catalyst exhibited only a 1.5 % weight loss after the reaction (Figure 6a), indicating minimal carbonaceous deposition.^[50] Desorption temperatures were observed at 177.4 °C and 238.6 °C (Figure 6b), both within the range of soft coke (165–400 °C),^[51] suggesting effective coke removal through heat treatment. Based on the highest desorption peak at 238.6 °C, a

calcination temperature of 300 °C was selected for catalyst regeneration, ensuring sustained performance in future cycles. The results, shown in Figure 6c, demonstrate excellent reusability of the catalyst, with conversion remaining high at 82.0 % for the first reuse and 80.0 % for the second. The product distribution exhibited minimal variation, with selectivity toward high-value liquid fuels maintaining significant levels at 79.2 % after the first reuse and 76.5 % after the second (Figure S10). These findings highlight the strong potential of the Ru–Pt/TiO₂ catalyst for repeated use in the tandem PE hydrogenolysis/methanol APR process, which is critical for practical implementation.

The Ru–Pt/TiO₂ bimetallic catalyst was employed in the tandem PE hydrogenolysis/methanol APR process, utilizing various plastic types such as high-density polyethylene (HDPE), HDPE bottles, commercial low-density polyethylene (LDPE), and a mixture of LDPE bags and PE, as shown in Figure 7. The catalyst achieved a conversion of 77.3 % with HDPE feedstock, alongside a selectivity of 75.0 % toward valuable liquid and lubricant oil products (Figure 7a). For HDPE bottles, the conversion was 75.3 %, with a 77.9 % selectivity for liquid and lubricant oils (Figure 7b). Commercial LDPE yielded a conversion of 71.6 % and a notably high selectivity of 88.9 % for the desired products (Figure 7c). Additionally, the reaction was conducted with a 1:1 mixture of LDPE bag and PE, as reported previously (Figure 7d).^[52] For this mixture, the conversion was 85.3 %, with a selectivity of 82.9 % for liquid and lubricant oils (Figure 7d). These results underscore the versatility of the Ru–Pt/TiO₂ catalyst in processing various plastic waste types into valuable products through the tandem PE hydrogenolysis/methanol APR process.

Interestingly, the product distribution and conversion in this tandem process were comparable to those achieved in PE hydrogenolysis reactions using an external hydrogen gas supply, despite the *in situ* hydrogen production from methanol APR in the PE hydrogenolysis process. This finding suggests that the tandem PE hydrogenolysis/methanol APR can be conducted as a single-step process, eliminating the need for grey hydrogen derived from petrochemical sources. Instead, hydrogen is generated *in situ* through methanol APR, utilizing the same catalyst and

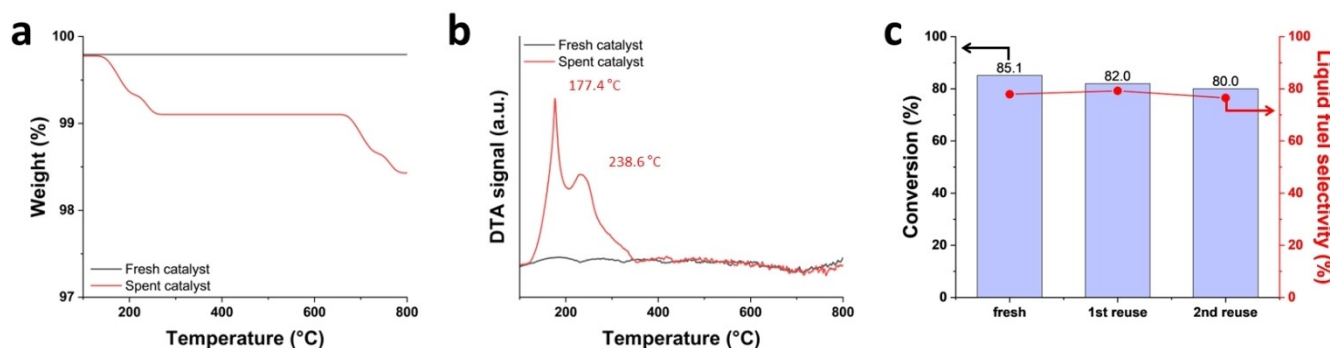


Figure 6. Characterization and reusability of spent catalyst in the tandem PE hydrogenolysis/methanol APR reaction. a. Thermogravimetric analysis (TGA) and b. Differential thermal analysis (DTA) of fresh and spent Ru–Pt/TiO₂ catalysts, heated to 800 °C at a rate of 10 °C/min under an airflow of 50 mL/min in an alumina cell. c. PE conversion as a function of reuse time, with detailed results provided in Table S15.

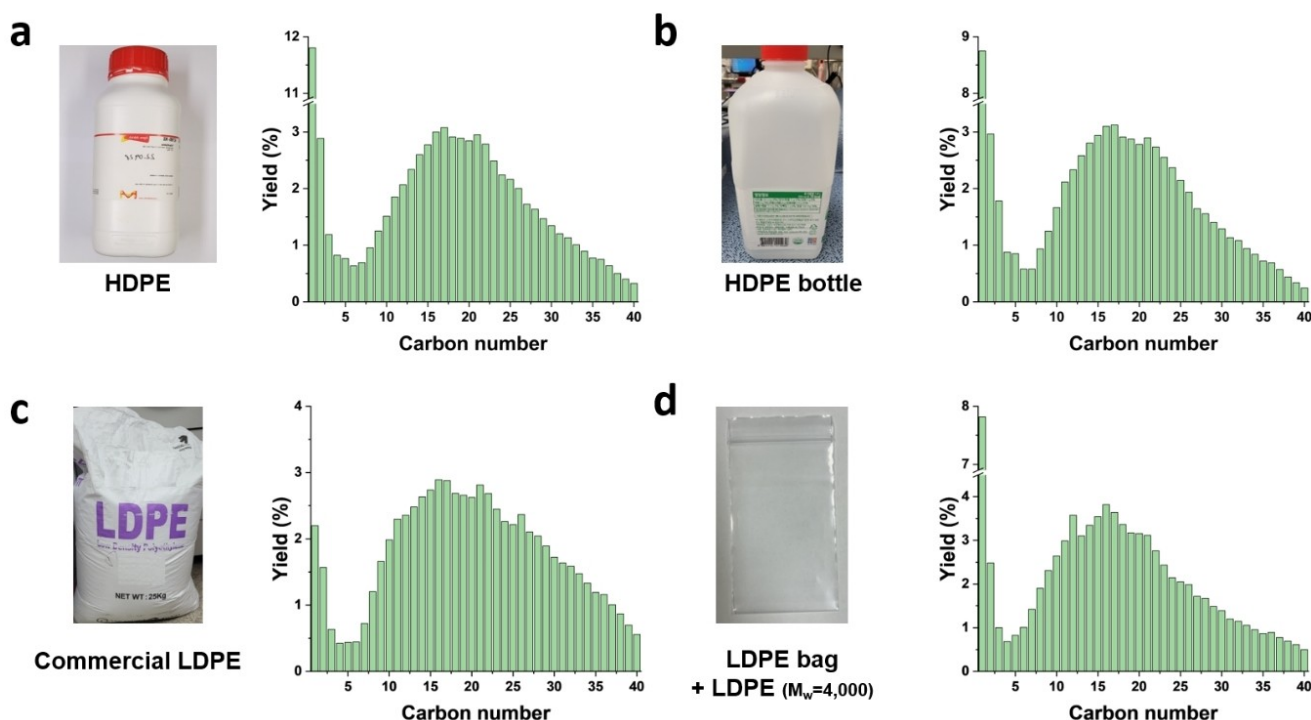


Figure 7. Tandem hydrogenolysis/methanol APR reaction results for post-consumer waste plastics. a. HDPE, b. HDPE bottle, c. Commercial LDPE, d. a mixture of LDPE bag and LDPE (M_n : ~1,700, M_w : ~4,000). Reaction conditions: Reactant 1 g (mixture: LDPE bag and LDPE were used 0.5 g, respectively), Ru–Pt/TiO₂ catalyst 0.1 g, 325 °C, 500 rpm, 12 h, 1 MPa Ar, methanol 0.5 g and water 1.5 g. Detailed results are shown in Table S16.

operating conditions as in the hydrogenolysis process. This integrated approach not only offers process intensification and energy savings but also aligns with the goals of sustainable green chemistry.^[15a,33] Additionally, the coupling of the endothermic methanol APR with the exothermic PE hydrogenolysis in a single-step reaction reduces the overall carbon footprint.^[40,53] Importantly, the one-pot tandem approach provides an environmentally friendly source of hydrogen for PE hydrogenolysis, significantly contributing to the sustainability of plastic waste depolymerization.

Conclusion

In summary, we successfully developed a tandem PE hydrogenolysis/methanol APR process as a sustainable alternative to conventional PE hydrogenolysis, which typically relies on an external hydrogen gas supply. After screening various catalysts, we found that the Ru–Pt/TiO₂ catalyst exhibited the highest activity, with Ru primarily responsible for PE hydrogenolysis and Pt facilitating methanol APR. Under optimal conditions, PE conversion reached 85.1 %, with 81.0 % selectivity toward liquid and lubricant oils. Notably, the product distribution was similar to that of conventional hydrogenolysis using external hydrogen, with products being predominantly n-alkanes, simplifying downstream separation. The tandem process also proved effective with real plastic waste, achieving conversions of 71.6 % to 85.3 % and selectivity toward liquid fuels and lubricant oils between 75.0 % and 88.9 %. This study introduces an unprecedented

catalytic plastic depolymerization process incorporating methanol APR with PE hydrogenolysis to reduce reliance on external hydrogen gas, presenting a promising route toward more sustainable plastic recycling.

Supporting Information

The experimental procedures, supportive table, and images are shown in Supporting Information. The authors have cited additional references within the Supporting Information.^[29c,54]

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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